

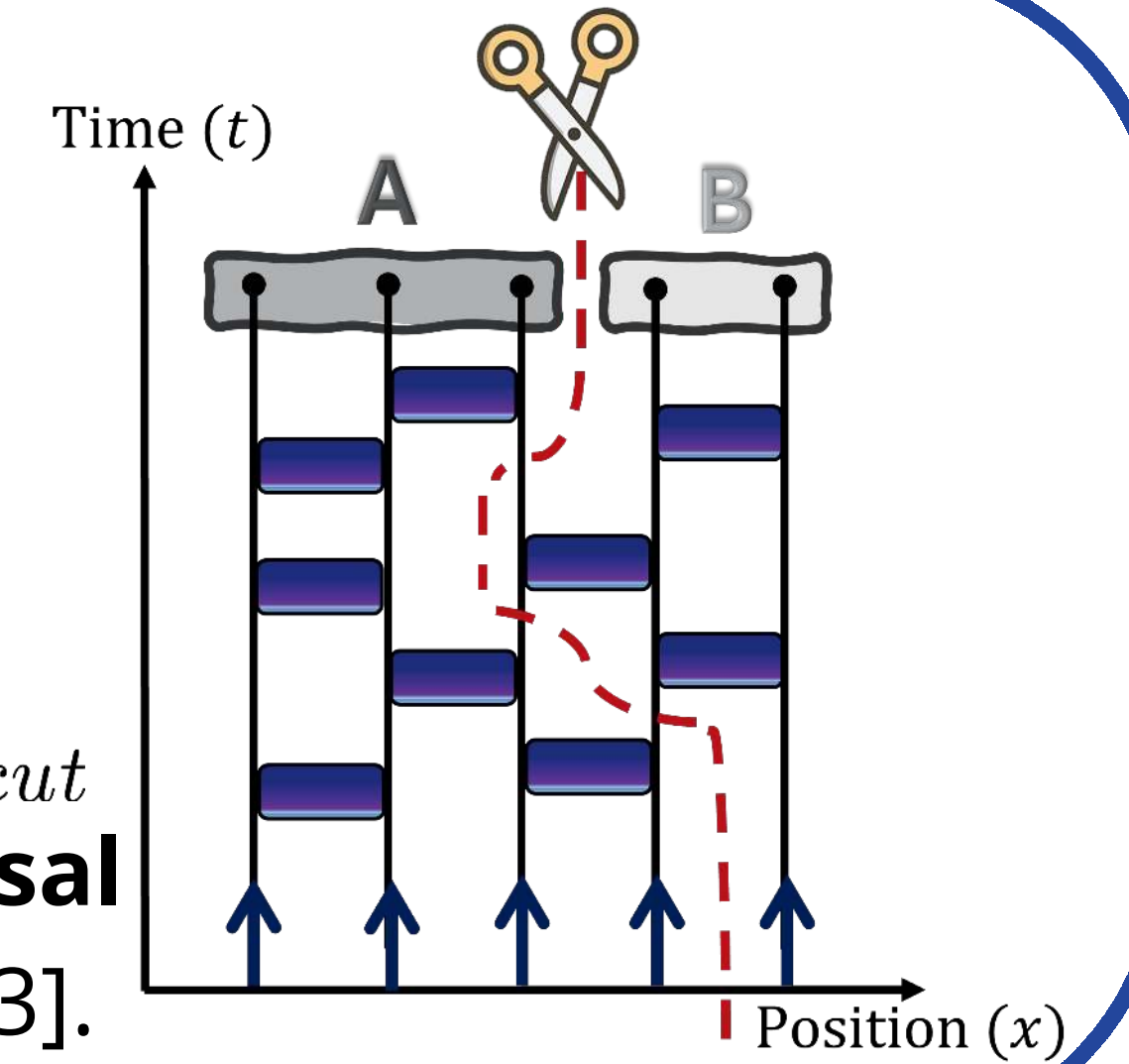
INTRODUCTION

Entanglement entropy is a highly nonlocal quantity, making it challenging to study its propagation over time.

$$S_\alpha = \frac{1}{1-\alpha} \log(\text{Tr}(\rho_A^\alpha))$$

To address this, **random quantum circuits** are studied as they offer a generic framework to understand the behavior of entanglement. In this context, the growth of entanglement is described by the **bipartite entropy** $S(x, t)$ [1,2].

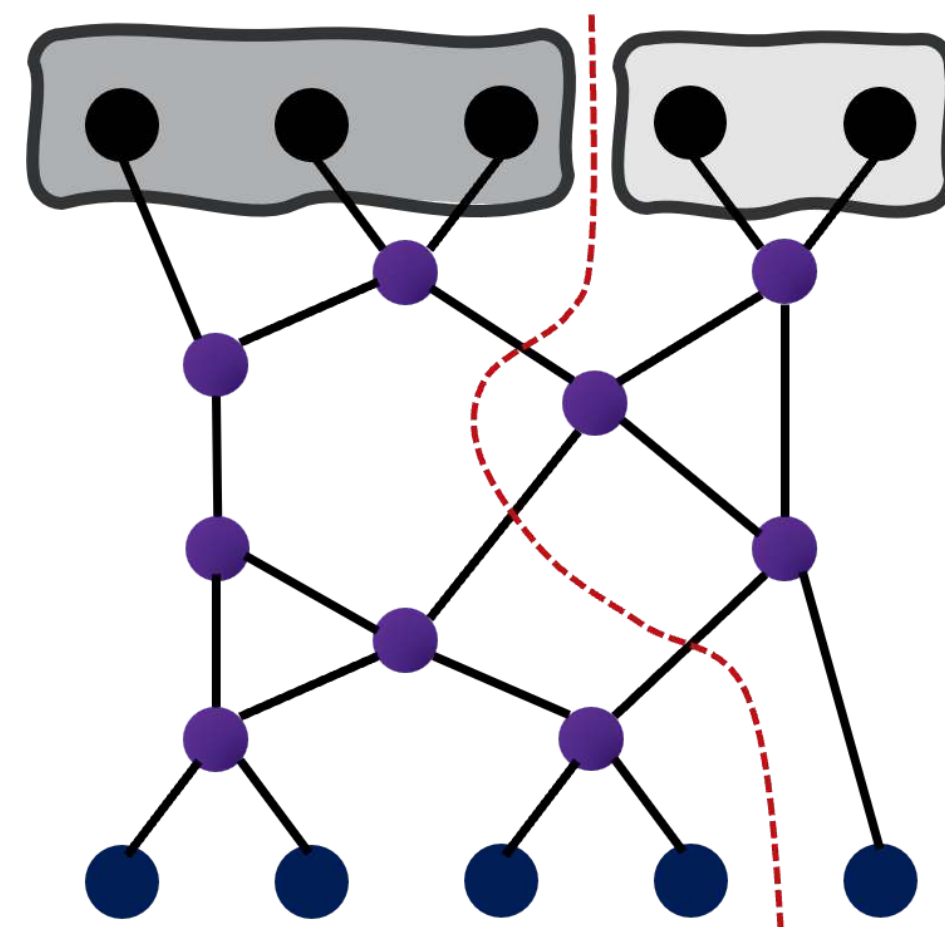
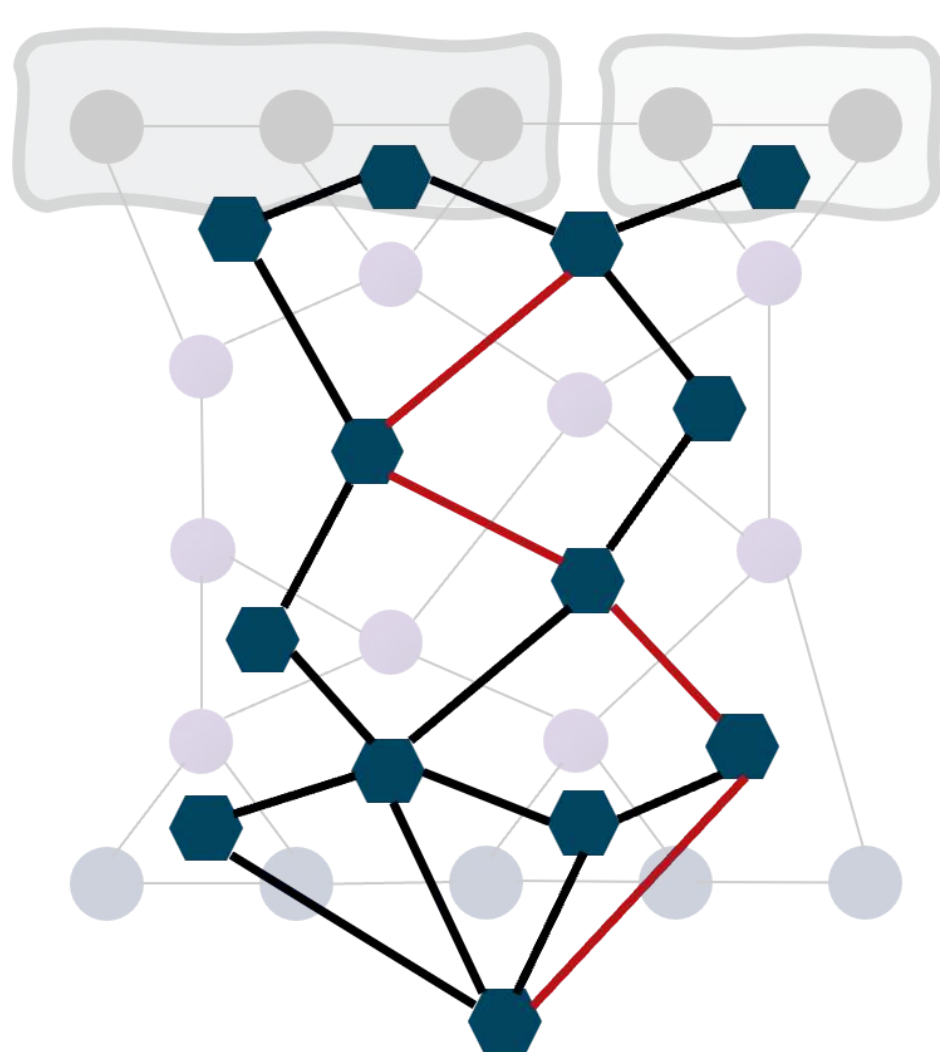
The **minimal cut** is a heuristic tool that connects entanglement entropy with network analysis. It represents a spacetime curve, dividing the system into two regions. While it provides only a lower bound for the entropy, $S_\alpha \leq N_{\text{cut}}$ it successfully captures the **universal properties** of entanglement dynamics [3].



METHODOLOGY

FIND THE MINIMAL CUT

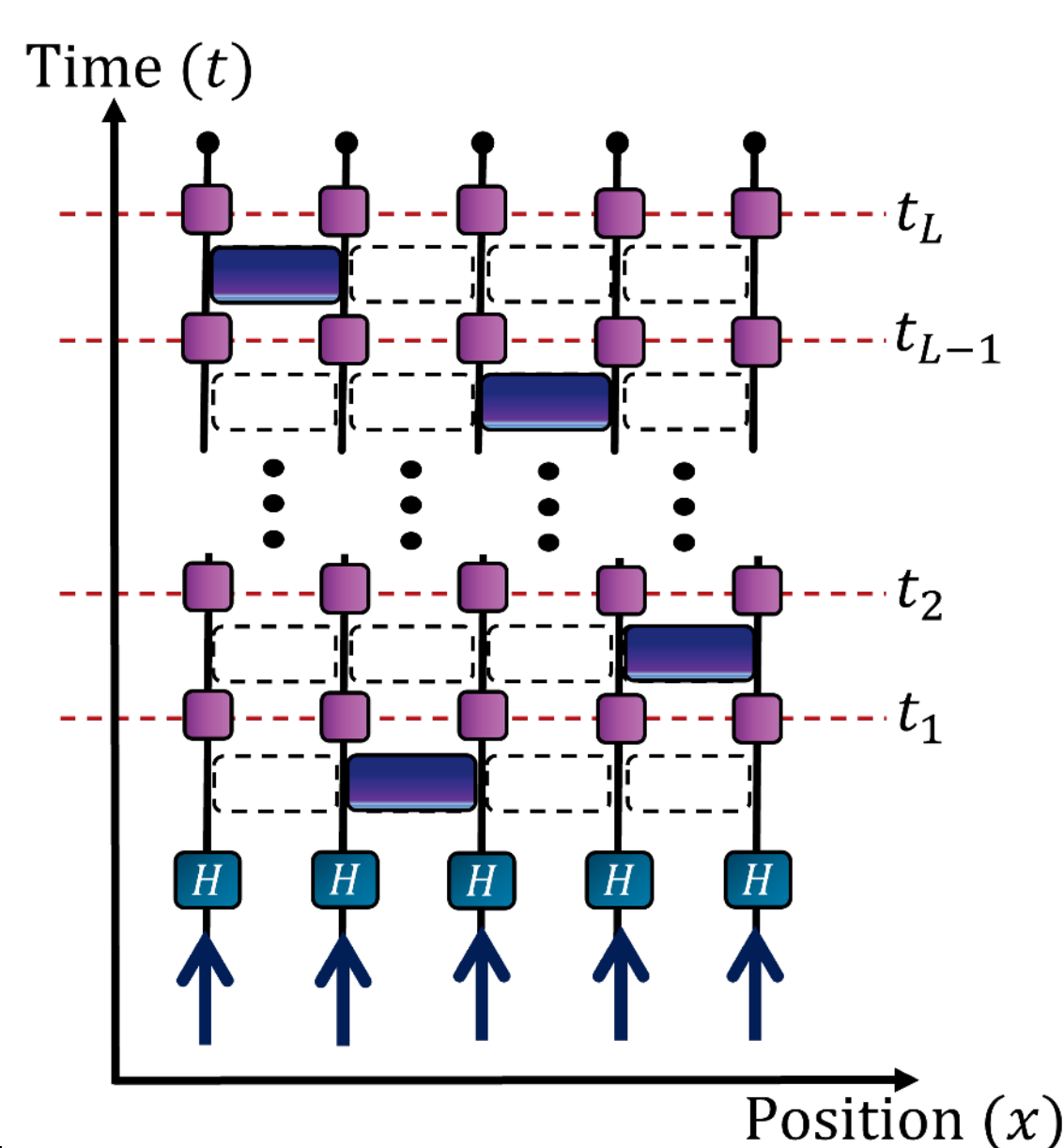
The problem is mapped onto a **graph** where nodes and edges represent circuit elements, with edge weights indicating the cost of dividing system regions.



An **algorithm** is developed to identify minimal cycles, map the flow to another graph, and determine the minimal path using **Dijkstra's algorithm**.

RANDOM CIRCUITS

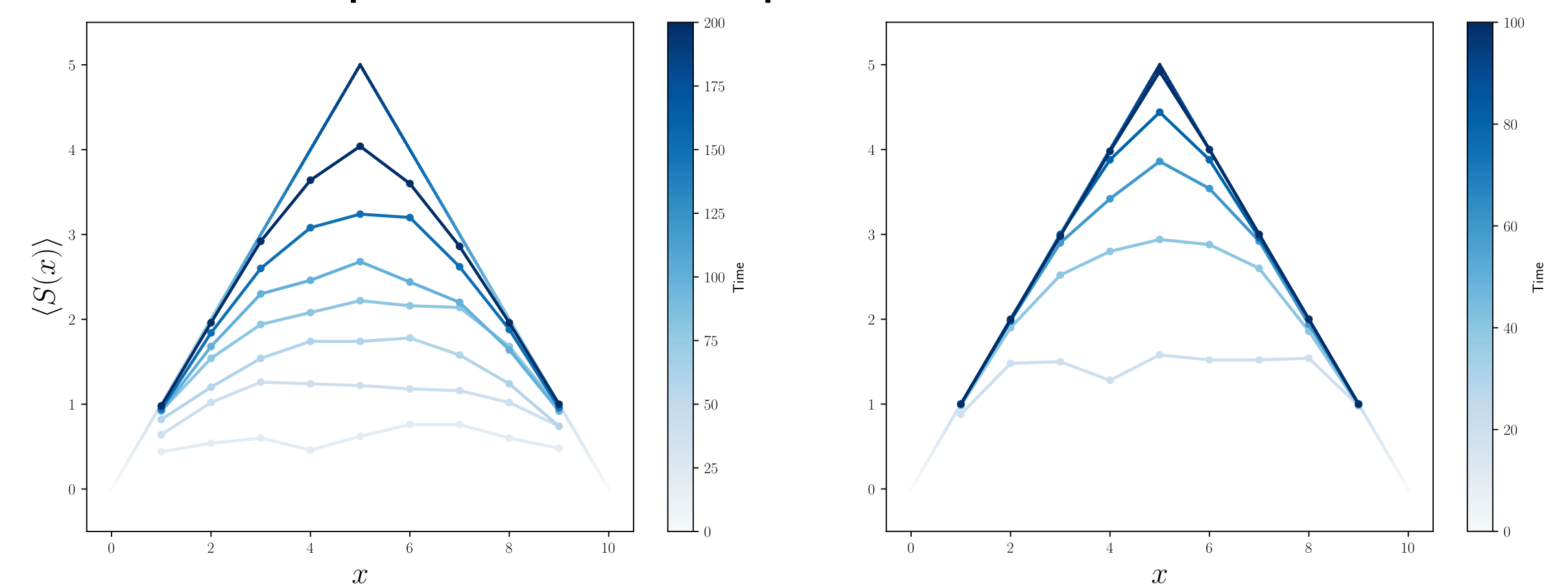
We analyzed the **time evolution** of random quantum circuits acting on spin-1/2 chains with open boundary conditions, starting from a product state with spins aligned in the positive z direction, and all qubits were initially placed in superposition using Hadamard gates.



At each time step t , a CNOT gate was applied between two qubits, with their positions selected **uniformly at random**. Between CNOT gates, single-qubit gates (Hadamard, Identity, or Phase) were applied randomly.

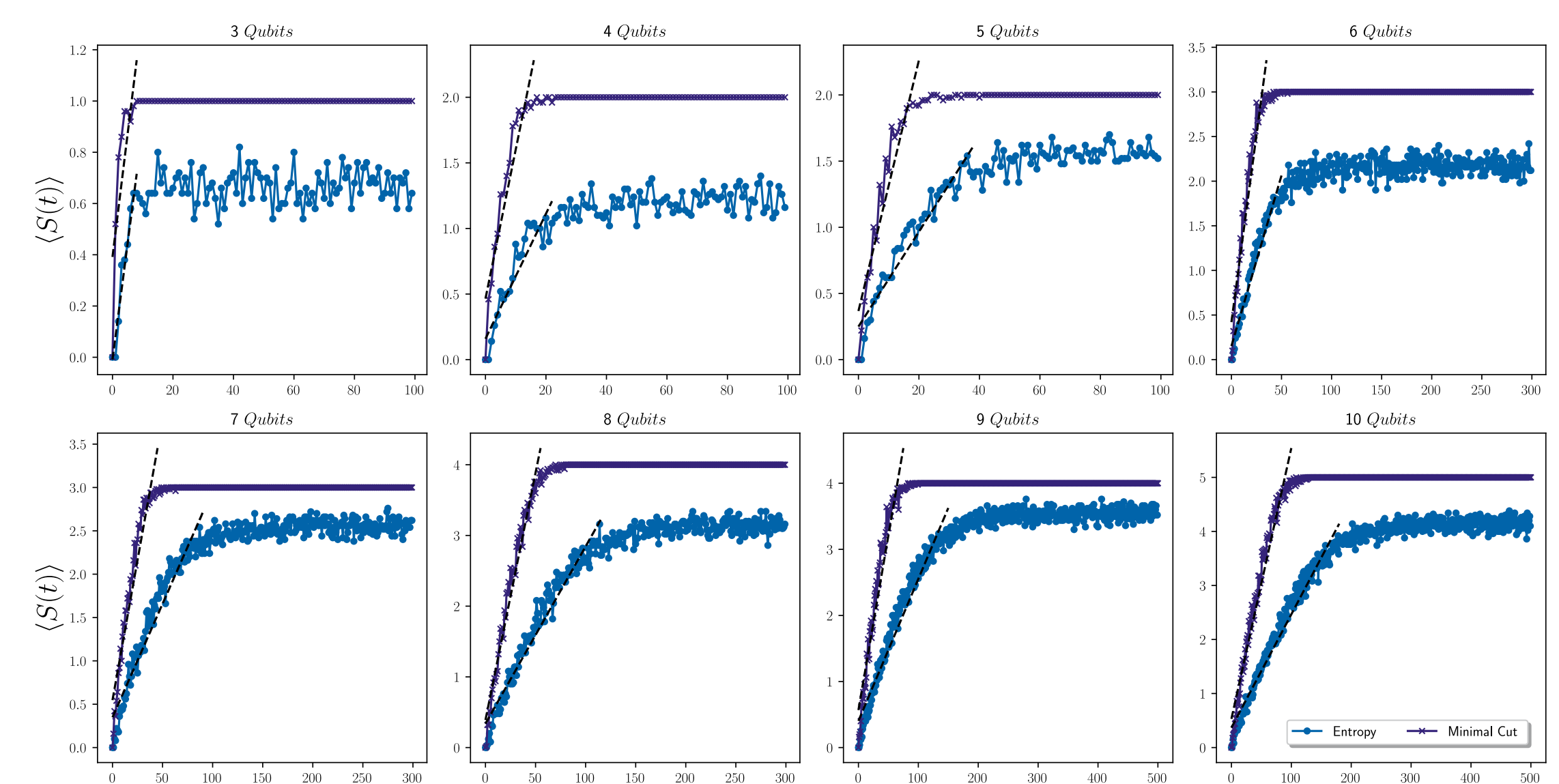
RESULTS

Entropy saturates in a pyramidal profile, bounded by $0 \leq S(x, t) \leq \log(D)$, where D is the smaller dimension of the Hilbert subspaces in the bipartite state.

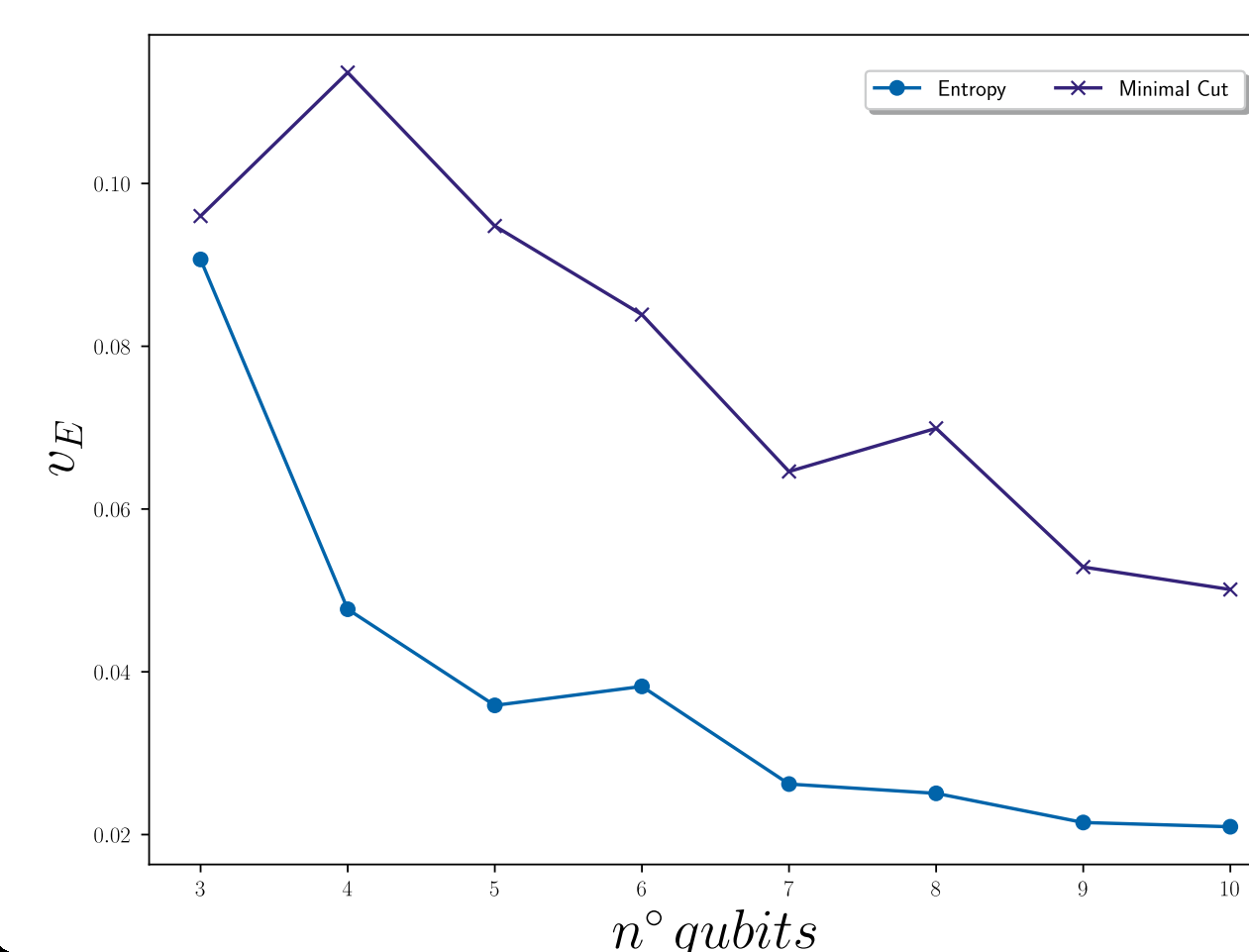


The average entropy over a set of 50 random circuits over time versus position for a 10-spin system. The left plot shows the real entropy, while the right plot shows the minimum cut estimate. The entropy is calculated for $\alpha = 1$, which corresponds to the von Neumann entropy.

In finite systems, $S(x, t)$ grows linearly with time, as $\langle S(x, t) \rangle = v_E t$, where v_E is the entanglement growth rate, until reaching a saturation point.



Average entropy for random circuits with qubit numbers ranging from 3 to 10, evaluated at $x = \frac{n}{2}$, where n is the total number of qubits. Each curve represents the mean over a set of 50 random circuits. Black dashed lines indicate linear regressions for the pre-saturation regime.



The entanglement rate decreases with the number of qubits. Consequently, achieving maximal entropy takes significantly longer in systems with more qubits.

CONCLUSIONS

The minimal cut saturates faster than the real entanglement entropy but exhibits the same growth behavior, providing a **computationally efficient** alternative to studying entanglement dynamics. The **saturation time** is $t \approx \frac{L}{2v_E}$, marking the system's thermalization point, erasing memory of the initial conditions. The simulation time does not allow observing the maximum entanglement in the real entropy calculations, as the growth in the saturation regime is very slow. For large qubit systems, this growth is modeled by the **KPZ equation**, which includes a term $t^{\frac{1}{3}}$ describing this regime.

REFERENCES

- [1] Nahum, A., Ruhman, J., Vijay, S., & Haah, J. (2017). Quantum entanglement growth under random unitary dynamics. *Physical Review X*, 7(3), 031016.
- [2] Fisher, M. P., Khemani, V., Nahum, A., & Vijay, S. (2023). Random quantum circuits. *Annual Review of Condensed Matter Physics*, 14(1), 335-379.
- [3] Skinner, B. (2023). Introduction to random unitary circuits and the measurement-induced entanglement phase transition. *arXiv preprint arXiv:2307.02986*.